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Optimal Variable, Lane Group–Based Speed Limits at Freeway Lane Drops: A Multiobjective Approach

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Abstract: This study develops optimal variable, *lane group–based*, speed limits for traffic control at freeway lane drop areas (e.g., work zones). The proposed approach adopts a simulation-optimization framework that utilizes a calibrated and validated macroscopic traffic flow model METANET, along with the microscopic traffic simulation model VISSIM, to develop the optimal speed limits. A multiobjective optimization framework is implemented whereby the model primarily seeks to improve traffic safety by reducing the average number of stops, while taking other objectives, such as the average travel time and throughput, into consideration. For optimization, the heuristic, biologically-inspired optimization technique known as particle swarm optimization (PSO), is utilized, and the ε-constraint method is adopted to allow for considering multiple objectives in the optimization process. The proposed traffic control strategy is then evaluated for a hypothetical freeway lane drop area under a real-world congested traffic scenario. The research findings show that the proposed lane group–based control strategy outperforms other *link-based*, variable speed limits reported in the literature. **DOI: 10.1061/JTEPBS.0000395.** © *2020 American Society of Civil Engineers.*

Author keywords: Freeway operations; Traffic simulation; Work zones; Variable speed limits; Particle swarm optimization (PSO).

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Introduction

Optimal traffic control for congested or oversaturated traffic conditions has always been of critical importance to mitigating the negative impacts of congestion on both traffic mobility and safety. On freeways, lane drop areas (e.g., work zones) are among the key locations where congestion occurs as a result of the reduction in the number of lanes and, hence, the available capacity. The resulting congestion not only affects travel time and throughput, but it also could increase the risk of highway accidents, and thus, could negatively impact the safety of operations.

The average number of stops per vehicle is one of the key measures of traffic safety, as well as mobility. The fewer the numbers of stops are, the smoother and safer the traffic flow is. This is because stop-and-go traffic flow situations, especially in congested traffic, lead to an increased risk of rear-end crashes (Golob et al. 2004). Crashes themselves could, in turn, lead to further deterioration in system performance as a result of the loss of additional roadway capacity because of the accidents and the closure of additional lanes. It also results in increased fuel consumption and emission rates (Barth and Boriboonsomsin 2009).

Variable speed limits (VSLs) on freeways, during congestion, are among the traffic control strategies that have attracted increased attention in recent years. From a safety perspective, the goal of the strategy may be to minimize the average number of stops per vehicle. On the other hand, for efficiency, the strategy may attempt to reduce travel time/delay, or to increase traffic throughput. In this study, we propose a traffic control strategy that estimates optimal VSLs, *on a lane-specific basis*, while considering both safety-related and efficiency-related, objectives. The strategy is considered useful for sites in which capacity is reduced as a result of lane drops and lane closures because of construction work or as a result of highway crashes. Furthermore, the strategy is designed to take advantage of the emerging technology of connected and autonomous vehicles (CAVs), which provides the opportunity to control vehicle speeds on a lane basis or even on an individual vehicle basis.

The key contributions of the current study include:

- Introducing a joint approach that utilizes both macroscopic and microscopic traffic flow models for developing optimal traffic control strategies. This is designed to speed up the optimization/search process, by first using the macroscopic model, METANET, to arrive at an approximate solution, which is then refined using the microscopic traffic simulation model, VISSIM version 5.40.
- Proposing an optimal, lane group–based, VSL traffic control strategy, and evaluating the strategy for a hypothetical freeway lane drop area under a real-world congested traffic scenario. Additionally, the new strategy is compared to two other control strategies (i.e., the traditional variable link-based speed limits and variable lane group–based speed limits with no right lane change allowed), and its superior performance is demonstrated.
- Presenting an optimization-simulation framework that utilizes the particle swarm optimization (PSO) algorithm for developing optimal traffic control strategies.
- Suggesting the ε-constraint method as the multiobjective optimization approach for minimizing the average number of stops per vehicle at congested freeway lane drop areas in order to improve traffic safety, while at the same time considering efficiency-related objectives, such as the average travel time and vehicle throughput.

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This paper is organized as follows. "Literature Review" section provides an overview of the key traffic flow models utilized by past studies of model-based traffic control. "Methodology" section presents the methodology followed in the study; specifically, this section describes: (1) the hypothetical problem considered in this study; (2) the traffic models utilized; (3) the adopted heuristic optimization technique; (4) the multicriteria optimization method; and (5) the benchmark traffic control scenario against which the proposed strategy is evaluated. The study's results are then provided in the "Results" section, which first reports on the calibration and validation of the METANET model, and then presents the main findings of the study regarding the evaluation of the proposed traffic control strategy. Finally, "Conclusions" section summarizes the conclusions of the study and offers some suggestions for future research.

Literature Review

Traffic Flow Models—A Brief Overview

Traffic flow models represent the foundation upon which traffic control strategies are developed. Models attempt to capture traffic dynamics, thereby allowing for the prediction ofthe impact of various candidate control strategies on the performance of the transportation system. Hoogendoorn and Bovy (2001) presented a comprehensive review of various traffic flow models. Traffic flow models are categorized according to their level of detail. On the one hand, microscopic traffic simulation models (such as Vissim, AIMSUN, etc.) strive to model individual driver behavior, and contain several parameters to mimic traffic dynamics in detail with high accuracy. On the other hand, macroscopic traffic flow models have fewer parameters and focus on representing aggregate traffic behavior for large networks using such variables as traffic density, mean speed, and flow.

The first macroscopic model Lighthill, Whitham and Richards (LWR) was introduced by Lighthill and Whitham (1955) and Richards (1956). In their study, the kinematic wave theory was used and traffic flow dynamics were expressed by means of a partial differential equation (i.e., the conservation equation). Later, the cell transmission model (CTM) was introduced, a first-order macroscopic traffic flow model that is a space-time discretized form of LWR model (Daganzo 1993, 1995). A higher-order or second-order macroscopic traffic flow model was introduced by Payne (1971). In addition to the partial differential equation regarding the conservation of vehicles, the Payne model contains a second partial differential equation expressing the speed dynamics or variations. Further, Messmer and Papageorgiou (1990) introduced the METANET model, a second-order macroscopic traffic flow model that is a space-time discretized and improved version of the Payne model. METANET and CTM are currently the two most widely used macroscopic traffic flow models. Both models are capable of reproducing traffic dynamics with an appropriate accuracy, but the METANET model displays better representation, especially of speed dynamics (Spiliopoulou et al. 2014).

Before the macroscopic flow models can be used in real-world applications, they must undergo a calibration and validation process to ensure that they are capable of accurately representing traffic dynamics. The nonlinear structure of those models makes performing the calibration process difficult, but thanks to heuristic optimization methods, it is possible to overcome this challenge. For instance, in the study of Spiliopoulou et al. (2015), three different optimization algorithms were utilized to calibrate METANET (namely the deterministic Nelder-Mead algorithm, the stochastic genetic algorithm, and the stochastic cross-entropy method). That study found that calibrating the model using the three different optimization algorithms yielded similar accuracy.

Freeway Lane Drops and VSLs

Bertini and Leal (2005) presented a good analysis of traffic flow behavior at bottlenecks such as freeway lane drop sites. More recently, Yuan et al. (2017) provided an in-depth investigation of the mechanism behind the capacity drop phenomenon at freeway bottlenecks; one of their main findings is that the stop-and-go waves negatively impact throughput to a greater extent than standing queues at the same site. To improve mobility and safety at such bottlenecks, different approaches of active traffic management (ATM) have been proposed (Brinckerhoff 2010; Neudorff and Mccabe 2015). VSLs are among the most effective ATM strategies that can be applied to bottlenecks and lane drop sites. A comprehensive overview of VSL control measures was covered in studies by (Lu and Shladover 2014; Khondaker and Kattan 2015b; Ma et al. 2016). Moreover, Katz et al. (2017) presented an exhaustive review of VSL operations, along with information gathered from interviews with a number of US. transportation agencies. These studies have shown that VSL systems, can not only improve traffic mobility, but can also lead to enhanced traffic safety. Specifically, the number and/or severity of crashes were, in general, found to decrease with the reduction of speed limits, as indicated in the study of Stuster et al. (1998).

In terms of setting appropriate *static* speed limits, there are various methodologies that are followed in real-world applications. These include the engineering approach, the expert systems, the optimization approach, and the injury minimization and safe system approach (Forbes et al. 2012). In this paper, we follow the optimization approach, but apply it to the VSL problem by attempting to estimate optimal speed limits that would maximize (or minimize) a single or multiple traffic performance criteria.

Optimal, Model-Based Traffic Control Strategies

Several model-based, optimal control strategies were proposed in previous research so as to enhance traffic performance on freeway networks. Papageorgiou et al. (2003) presented a review of traffic control strategies carried out in urban and freeway networks. In the study of Kotsialos et al. (2002), the researchers explored the optimal integration between route guidance and ramp metering on motorway networks based on the METANET model. Papageorgiou et al. (2008) studied the VSLs impact on aggregate traffic flow behavior. Hegyi et al. (2005) and Carlson et al. (2010) investigated the optimal coordination of VSLs and ramp metering on freeway networks using the METANET model. VISSIM was used in the study of Yang et al. (2009) to evaluate a lane-based traffic signal system to regulate merge at freeway work zones. AIMSUN with a local ramp metering strategy ALINEA were utilized in the study of Lentzakis et al. (2008), in which they presented a real time traffic signal–based merge control strategy, upstream of the work zone, for throughput maximization. In this paper, we suggest a joint approach that utilizes both macroscopic and microscopic traffic flow models for developing optimal traffic control strategies.

Multiobjective Optimization

In terms of the objective of traffic control, the adoption of a multiobjective framework has its advantages over the use of a single objective, because it allows for considering the trade-offs among several objectives that are often conflicting with one another. To date, most of the prior optimization-based traffic flow control studies on *freeways* were based on single-objective problem formulations.

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Some notable exceptions though, include studies such as (Yang and Wen 2010; Khondaker and Kattan 2015a; Wen 2015; Li et al. 2016; Yu and Fan 2018). These studies proposed multicriteria traffic control strategies, implemented at the link level, for ramp metering, route guidance, or VSLs. Our current work attempts to contribute to this growing research area in the literature as it takes a further step to propose VSL control strategies, *at the lane-group level*, for freeway lane drop sites, while considering multiple objectives. Before proceeding, however, it is worth mentioning that the literature on *traffic signal control* appears to have considered multiobjective formulations more often than the freeway control literature. Examples include studies by (Stevanovic et al. 2009; Li et al. 2013; Dujardin et al. 2015; Long et al. 2015; Stevanovic et al. 2015; Li and Sun 2018). Signal control is not the focus of this paper.

Traffic Control Considering Vehicle Connectivity and Automation

In recent years, there has been an unprecedented interest in researching traffic control strategies that consider, as well as leverage, vehicle connectivity and automation. (Roncoli et al. 2015a, b) introduced a first-order multilane traffic flow model for motorways, and then applied it to develop a traffic control strategy aimed at regulating speed limits, lane changing, and ramp metering in the presence of vehicle automation and communication systems. Ntousakis et al. (2016) developed a trajectory planning technique to create a smooth merging process. Han et al. (2017) proposed VSL control strategies using connected vehicle technology to improve bottleneck discharge rate at freeways. Letter and Elefteriadou (2017) developed strategies based on connected vehicle technology using microscopic traffic simulator (CORSIM). A multiclass microscopic model, for heterogeneous platoon consisting of both human-driven vehicles and CAVs, was proposed in the theoretical and numerical study of Jia et al. (2019). In an interesting case study with more than twenty human-driven vehicles and only one autonomous vehicle, Stern et al. (2018) showed that the low penetration rate of the autonomous vehicles has promising benefits on regulating overall traffic flow and making it smoother. According to their experiments, significant improvements were achieved in the traffic flow measures (namely the velocity standard deviation, fuel consumption, braking events, and throughput).

However, one of the downsides of those traffic control strategies that are based exclusively on CAVs is that they cannot be implemented in the short-term, but have to wait for the mainstream adoption of such technologies by drivers. Such wide-scale adoption requires addressing several challenges in terms of concerns about safety, security, privacy, policies, unemployment rate, cost, etc. (Bagloee et al. 2016; Litman 2018). Our proposed approach in this article can be implemented in a traditional-traffic environment, while having the flexibility to be adopted for the futuristic connected-and-automated environment as well.

Building on the previous studies previously outlined, the current study describes a framework for the development of optimal variable *lane group–based* speed limits at freeway lane drop areas, which considers multiple objectives and utilizes a joint modeling approach with both microscopic and macroscopic modeling.

Methodology

The methodology adopted in the current study employs both macroscopic and microscopic traffic flow models to develop optimal traffic control strategies. The idea is to derive preliminary solutions to the optimization problem first using the macroscopic model; the advantage here stems from the fact that a macroscopic model requires much less time to run compared to a microscopic model. Those solutions are then used as a start point for another optimization problem that is based upon the more detailed microscopic model. Fig. 1 displays the flowchart that shows the steps followed to estimate the optimal variable lane group–based speed limits. First, after calibrating and validating the macroscopic traffic flow model (METANET), we use it to analytically estimate the optimal VSLs, as shown in the left part of the flowchart (the part of the flow chart enclosed by the dashed line box in Fig. 1). After that, as shown in the right part of the flowchart, the microscopic traffic simulation model (VISSIM) is utilized, whereby the solutions derived using METANET are used as a start point (i.e., by seeding the initial population of the heuristic optimizer, the PSO, with the solution found using METANET; we refer to this as the METANET agent). In addition to the METANET agent, we also seed the initial population of PSO with agents, corresponding to the case of no control case (i.e., when no control action is implemented); we call this agent the BASE agent. Planting the BASE agent in the initial population guarantees that the optimization process only improves on the status quo. Optimization, whether based upon the macroscopic or microscopic model, is implemented within a multiobjective framework via the ε-constraint method.

The microscopic model we used is VISSIM (version 5.40) (PTV 2012b), one of the most widely used traffic simulation models at the present time. To integrate VISSIM within the optimization framework, its component object model (VISSIM-COM) (PTV 2012a) was used to access and control the various attributes of the VISSIM model. MATLAB (version R2015a) was used as the hosting programming language environment. The following section gives more details about the VISSIM model, the METANET model, the geometric layout of the case study utilized, and the control strategies developed and evaluated in the current research.

Microscopic Traffic Simulation Model (VISSIM)

VISSIM is a microscopic traffic flow model widely used for traffic network representation and simulation. VISSIM was selected in this study for the detailed (i.e., microscopic) modeling of the impact of VSLs, implemented at the *lane level* (and not just the link level), at freeway lane drop areas. The following subsections present a description of the freeway lane drop site considered in this study and modeled in VISSIM.

Layout of the Freeway Lane Drop Area Modeled

A hypothetical freeway lane drop area was created in VISSIM [Fig. 2(a)]. The upstream link consists of five lanes (L1, L2, ..., L5) and its length is 10 km. It was discretized into 10 sections or segments (S1, S2, ..., S10), each 1 km in length; control will be exerted along the length of this upstream link. The downstream link, on the other hand, consists of two lanes and its length is 1 km. The two sections are connected in VISSIM using a connector consisting of two lanes, and which is 0.1 km long. Lane width was fixed at 3.5 m. It was assumed that there were no obstructions upstream the lane drop, and that traffic approaches the modeled section with a speed equal to the free-flow speed.

In order to model loop detectors or sensors, groups of data collection points (D0, D1, ..., D10) were installed on the controlled part, as shown in Fig. 2(a) (displayed as rectangles). Each group consisted of five data collection points corresponding to the five lanes. D0 was installed at the entrance of the upstream link to record the input demand and mean speed, while D1 through D10 were installed in the middle of each section for the purpose of getting the traffic density and mean speed of each section. In order to model the VSLs, desired speed decisions were installed at five locations (U_1, U_2, ..., U_5) over the controlled part, as shown in Fig. 2(b) (displayed as circles).

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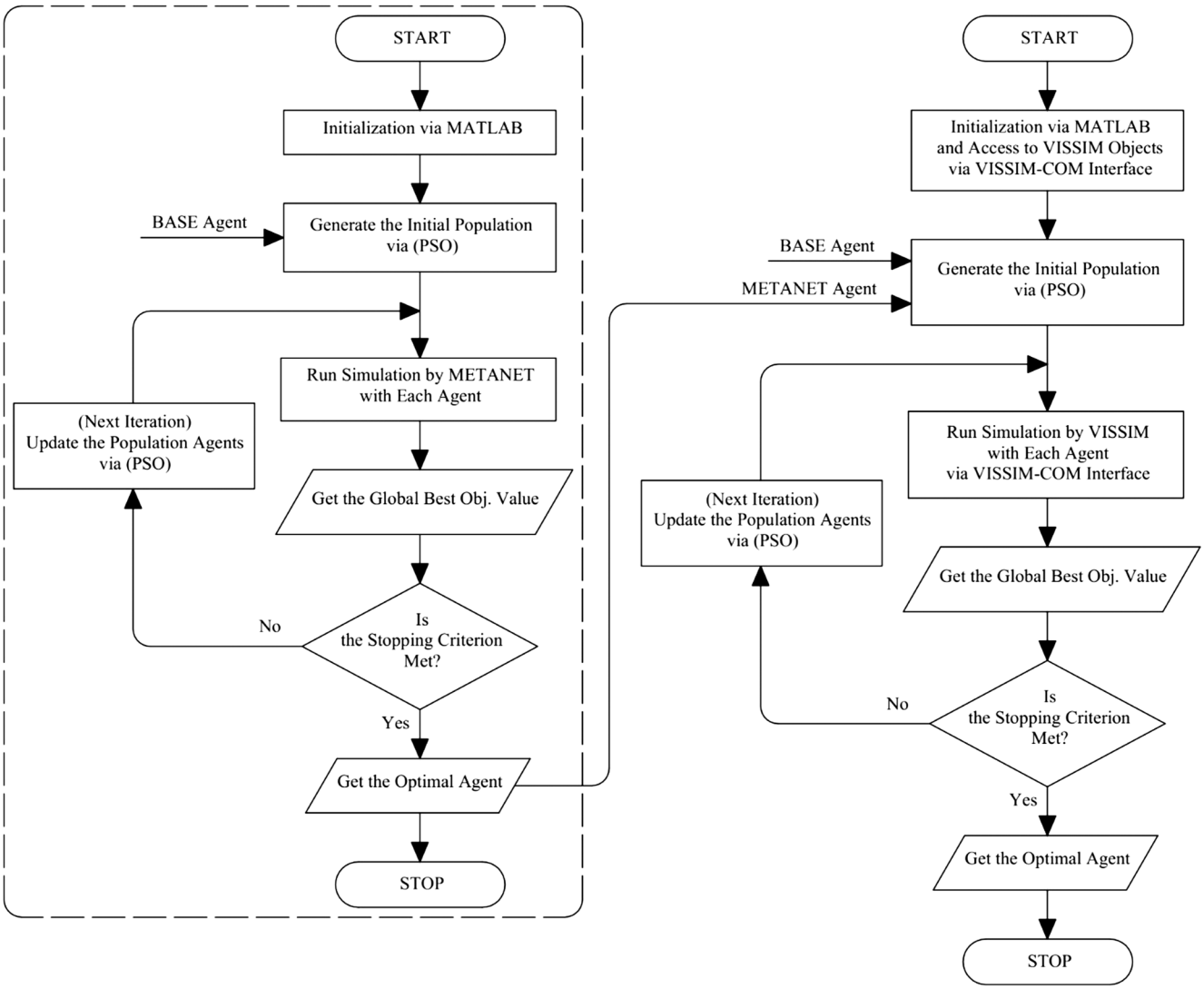


Fig. 1. Proposed control strategy scheme consisting of using both the macroscopic and microscopic traffic flow models jointly.

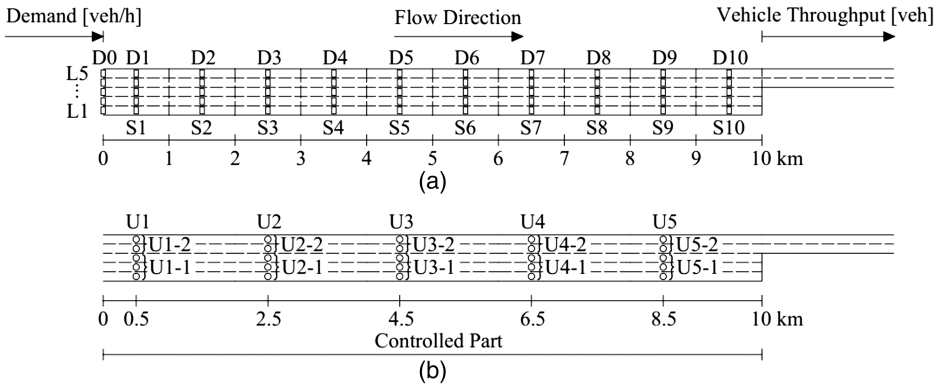


Fig. 2. (a) The freeway lane drop area layout; and (b) the speed limits installation over the controlled part.

Base Case (No Control Scenario)

In order to mimic a real-life congested traffic scenario in VISSIM, vehicle inputs were set as follows. At the beginning of the simulation horizon, a light traffic demand equal to 3,000 vehicles (veh)/h enters the upstream link between time step 0 and 240 s; this is well below the available capacity of even the 2-lane section after the lane drop. Following this initial demand period, a higher demand equal to 6,000 veh/h lasts for one full hour, from time step 240 to 3,840 s; this heavier demand level is well above the available capacity of the 2-lane section so as to cause congestion upstream of

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the lane drop. Vehicle types were assumed to be only passenger vehicles. Desired speed for all vehicles was assumed to be 100 km/h. The Wiedemann 99 traffic model, used within VISSIM, was assumed to capture driver behavior. All VISSIM parameters were set to the software's default values unless otherwise stated.

Macroscopic Traffic Flow Model (METANET)

The following subsections present a description of the structure of the METANET model introduced by Messmer and Papageorgiou (1990) and the calibration and validation process that was followed in this study to ensure that METANET is replicating as much as possible the detailed vehicle behavior modeled in VISSIM.

Model Structure

The METANET model is a second-order space-time discretized macroscopic traffic flow model. It discretizes each freeway link into a number of sections or segments (where the segment index $i = 1, 2, \ldots, I$ and I is the total number of segments). Each segment length (L_i) is assumed herein to be equal to 1 km; note that the roadway geometric characteristics [e.g., the number of lanes (λ_i)] within each segment need to be homogenous. The total simulation time horizon is discretized into uniform time steps (T) which were set to be equal to 10 s, in agreement with the relation ($T \leq L_i/V_{free}$), where V_{free} is the free-flow speed, and which is assumed to be equal to 100 km/h in our case. At each time step k (where k is the simulation time index = 0, 1, 2, $\ldots$, K and K is the total number of time steps), the METANET model computes the aggregate traffic variables for each segment, namely the traffic density $\rho_i(k)$, the mean speed $v_i(k)$, and the exit flow $q_i(k)$ according to the following equations:

$$\rho_i(k+1) = \rho_i(k) + \frac{T}{L_i \cdot \lambda_i}(q_{i-1}(k) - q_i(k)) \tag{1}$$

$$v_i(k+1) = v_i(k) + \frac{T}{\tau}(V(\rho_i(k)) - v_i(k)) + \frac{T}{L_i} v_i(k)(v_{i-1}(k) - v_i(k)) - \frac{\theta \cdot T}{\tau \cdot L_i}\left(\frac{\rho_{i+1}(k) - \rho_i(k)}{\rho_i(k) + \psi}\right) \tag{2}$$

$$q_i(k+1) = \rho_i(k+1) \cdot v_i(k+1) \cdot \lambda_i \tag{3}$$

where θ, τ, and ψ = model parameters; and $V(\rho_i(k))$ = desired speed (corresponding to the fundamental diagram), computed as

$$V(\rho_i(k)) = V_{free} \cdot exp\left(-\frac{1}{a}\left(\frac{\rho_i(k)}{\rho_{crit}}\right)^a\right) \tag{4}$$

where a = a parameter; and ρ_{crit} = critical density [assumed to be equal to 33.5 veh/km/lane (ln)]. Note that at a lane drop segment, the following speed reduction term is added to the mean speed relation [Eq. (2)]

$$-\frac{\phi \cdot T \cdot \Delta\lambda \cdot \rho_i(k) \cdot v_i(k)^2}{L_i \cdot \lambda_i \cdot \rho_{crit}}$$

where ϕ = a parameter; and $\Delta\lambda$ = number of dropped lanes.

It is worth noting that in the event of using METANET for freeway traffic control applications such as the development of VSLs, the speed limit [$v_{limit,i}(k)$], in agreement with Hegyi et al. (2005), is modeled as follows:

$$V(\rho_i(k)) = min\left(V_{free} \cdot exp\left(-\frac{1}{a}\left(\frac{\rho_i(k)}{\rho_{crit}}\right)^a\right), v_{limit,i}(k)\right) \tag{5}$$

Model Calibration and Validation

The METANET model must undergo a calibration and then a validation operation before use to make sure it can accurately replicate traffic dynamics. This is done by estimating the optimal values for the METANET model parameters that would enable the model to capture traffic dynamics accurately. In this study, the PSO algorithm (see subsection "Heuristic Optimization Technique (PSO) and the Multiobjective Approach (ε–constraint Method)" for more details about PSO), was utilized in the model calibration step (as mentioned before, PSO is also used to find the optimal VSLs themselves). PSO was chosen because of its capability to solve nondifferential and nonlinear optimization problems, such as the problem of calibrating the METANET model.

For calibration, the objective function is designed to minimize the root mean squared error (RMSE) between the set of values for traffic densities $\rho_i(k)^V$ and mean speeds $v_i(k)^V$ of the segments, as derived from the VISSIM model (i.e., data collected from D1, D2, $\ldots$, D10), and those values [$\rho_i(k)^M$ and $v_i(k)^M$] calculated via METANET for all time steps [Eq. (6)]. In other words, our goal here is to make METANET results close to those derived from VISSIM, since our framework is utilizing METANET as an approximation of VISSIM, which has the advantage of requiring significantly less time to run compared to VISSIM. The decision variables of the optimization model to be solved by PSO are the METANET model parameters (a, τ, θ, ψ, and ϕ) plus the downstream flow capacity $\bar{Q}_{cap}$ and downstream maximum density $\bar{\rho}_{max}$ downstream the lane drop; these were included since they are necessary for the exit boundary conditions ($q_I(k) \leq \bar{Q}_{cap}$), ($\rho_{I+1}(k) \leq \bar{\rho}_{max}$). By estimating those specific seven variables, our resulting METANET model would be ready for use; only the demand and mean speed corresponding to the considered traffic conditions will be required as the input variables. The minimization of the objective function is subject to the METANET model structure and nonnegativity constraints. Note that the estimated model parameters are assumed herein to apply to all the freeway segments

$$min\,\text{RMSE} = \sqrt{\frac{\sum_{k=0}^{K}\sum_{i=1}^{I}(\rho_i(k)^V - \rho_i(k)^M)^2}{K \cdot T}} + \sqrt{\frac{\sum_{k=0}^{K}\sum_{i=1}^{I}(v_i(k)^V - v_i(k))^2}{K \cdot T}} \tag{6}$$

To summarize, the steps of the algorithm implemented to calibrate METANET model parameters were as follows:

1. START and INITIALIZE via MATLAB.
2. GENERATE an initial, random population via PSO method.
3. Run the simulation model and calculate the value of the objective function, Eq. (6), for each agent (i.e., for each set of values for the METANET model's parameters a, τ, θ, ψ, and ϕ, plus $\bar{Q}_{cap}$ and $\bar{\rho}_{max}$).
4. GET the global best objective value.
5. IF the stopping criterion is met, GET the optimal agent and STOP. OTHERWISE, UPDATE the population agents via PSO method and then REPEAT Steps 3 to 5.

Following calibration, the ability of the calibrated METANET model to replicate traffic dynamics was checked or validated on a different traffic scenario other than that considered for model calibration. This is described in more detail in the subsection "METANET Model Calibration and Validation Outcomes."

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It should be noted that, according to our proposed approach, METANET is used to derive *link-based* VSLs, which would then serve as a good start point for searching for the optimal *lane group–based* VSLs using VISSIM as the evaluation function. Using the METANET agent as a candidate solution in the initial population of PSO method is a better alternative compared to starting the optimization from some bad randomly-generated agent.

Heuristic Optimization Technique (PSO) and the Multiobjective Approach (ε–constraint Method)

As previously mentioned, PSO is among the most well-known of nature-inspired optimization techniques that have been widely utilized in solving nonlinear optimization problems. The algorithm, which was originally introduced by Kennedy and Eberhart (1995), mimics the social behavior of bird swarms in search for food. PSO was selected in this study because it consumes less computation time and has fewer parameters and operators compared to other heuristic optimization methods like genetic algorithms (Hassan et al. 2005).

PSO proceeds in a number of iterations. In every iteration (n), the position of each particle (o), denoted by X_o, is calculated from the position equation [Eq. (7)], and its velocity (Υ_o) is computed from the velocity equation [Eq. (8)]. In these equations, P_o represents the particle best solution that has been achieved so far (cognitive knowledge), G refers to the global best solution that has been found among all the swarm particles so far (social knowledge), w is the inertia coefficient,(c_1 and c_2 are the acceleration coefficients, and r_1 and r_2 are uniformly distributed random numbers in the range [0, 1]. The PSO technique runs until the stopping criterion is met

$$X_o^{n+1} = X_o^n + \Upsilon_o^{n+1} \tag{7}$$

$$\Upsilon_o^{n+1} = w \times \Upsilon_o^n + c_1 \times r_1 \times (P_o^n - X_o^n) + c_2 \times r_2 \times (G^n - X_o^n) \tag{8}$$

The constriction factors introduced in (Clerc and Kennedy 2002) were adopted herein. The inertia coefficient (w) is set to be equal to the constriction factor ($\mathbb{C}$), and the acceleration coefficients are set as $c_1 = \mathbb{C}\Delta_1$ and $c_2 = \mathbb{C}\Delta_2$. The constriction factor is computed as $\mathbb{C} = \frac{2}{|2-\Delta-\sqrt{\Delta^2-4\Delta}|}$, where the control parameter $\Delta = \Delta_1 + \Delta_2 > 4$. The value of Δ_1 and Δ_2 is set to 2.05 each.

It is worth pointing out that (Coello and Lechuga 2002) first introduced an extension to the PSO to solve multiobjective optimization problems, by means of the Pareto dominance technique, to define the search direction of each particle. An extensive review of the different proposals presented for multiobjective particle swarm optimization (MOPSO) was covered in (Reyes-Sierra and Coello 2006). In our methodology, in order to prevent significant reductions in the value of the other traffic performance measures, the ε-constraint method was adopted (Ehrgott 2005). It was chosen herein for its simplicity and appropriateness for the present study's assumptions and goals. In this technique, only one of the objectives, which is the most desired one, is selected to be minimized while the other objectives are converted into constraints [Eq. (9)]

$$\min \boldsymbol{f}_j(\boldsymbol{v}) \tag{9}$$

$$subject\ to \boldsymbol{f}_t(\boldsymbol{v}) \le \boldsymbol{\varepsilon}_t$$

where f = vector of objective functions ($\boldsymbol{f} = f_1, f_2, \ldots, f_F$); the subscripts j and t = indices of the objectives ($t = 1, 2, \ldots, F$, and $t \neq j$), where F is the total number of objective functions; $\boldsymbol{v}$ = vector of decision variables ($\boldsymbol{v} = v_1, v_2, \ldots, v_m$), where m is the total number of decision variables; and $\boldsymbol{\varepsilon}$ = vector of upper bounds. The vector of upper bounds, $\boldsymbol{\varepsilon}$, can be determined from the base scenario as $\boldsymbol{\varepsilon}_t = \mathbb{Z}_t \cdot \boldsymbol{f}_t^{\text{base}}$, where $\mathbb{Z}$ is a vector of the predefined percentage values with regard to the corresponding values of those objective functions obtained from the base case $\boldsymbol{f}_t^{\text{base}}$. For example, a value of $\mathbb{Z}_t$ equal to 95% would mean that one is willing to accept only up to a 5% deterioration in the value of that particular objective function.

Traffic Control Strategies

This subsection will describe the different traffic control strategies that were considered in the present study, in addition to the no control case, Case 0, of the subsection "Base Case (No Control Scenario)."

Case 1: Variable *Lane Group–Based* Speed Limits Control Strategy

The increased levels of connectivity and automation within the transportation system, a trend that is likely to continue and accelerate in the future, afford the opportunity to develop VSLs on a *lane group-specific* basis (and not just on a link-specific basis). In fact, in the future, it would be quite possible to control the speeds and trajectories of individual vehicles. Given this, the strategy we focus on in this study is to regulate speed limits on a *lane group*-specific basis. The layout of the case study is shown in Fig. 2(a), whereas the locations in which the speed limits are to be controlled are shown in Fig. 2(b) [$U_1, U_2, \ldots, U_5$ (represented by circles)]. These positions are distributed in such a way so as to cover the traffic flow trajectory over the whole controlled part. The specific traffic scenario for the base case was described in the subsection "Base Case (No Control Scenario)." However, it is noted that the proposed framework is generic in nature, and can be applied to any other real-world case that differs in the layout of the site (i.e., the number of lanes upstream and downstream the lane drop).

To avoid the occurrence of instabilities in traffic flows, which could happen if the difference among the speed limits of the different lanes at the same position was large, the lanes at the same position were classified into two groups according to whether they were closed (those that are discontinued at the lane drop) or open lanes (those which continue beyond the lane drop). As shown in Fig. 2(b), the first group, the closed lanes, includes the three right lanes L1, L2, and L3, whereas the open lanes group includes the two left, Lanes L4 and L5. At the same position, the values of the speed limits are kept the same across the lanes belonging to the same group; hence the name *lane group–based VSLs*. For instance, the variables $U_{3\text{-}1}$ and $U_{3\text{-}2}$ represent the value of speed limits for the lanes at the same Position 3, for the first group (i.e., the closed lanes) and the second group (i.e., the open lanes), respectively. The controlled time window, when the speed limits are controlled, is assumed to extend from the moment when the high demand starts (Cs), until the moment when the demand returns back to its low level (Ce). Note that in the case of light traffic, there is no need for active traffic control and, therefore, the proposed control strategy is deactivated and speed limits return to their static, posted levels.

For optimization, the main objective function to be minimized $f_1(\boldsymbol{v})$ is expressed as the average number of stops per vehicle in order to improve traffic safety. With this, the multiobjective optimization framework constrains two additional objectives $f_2(\boldsymbol{v})$ and $f_3(\boldsymbol{v})$, namely the average travel time and the vehicle throughput, respectively, using the ε-constraint method according to Eq. (9) previously mentioned.

When using VISSIM, the values of the three objective functions previously mentioned can be easily obtained from the available

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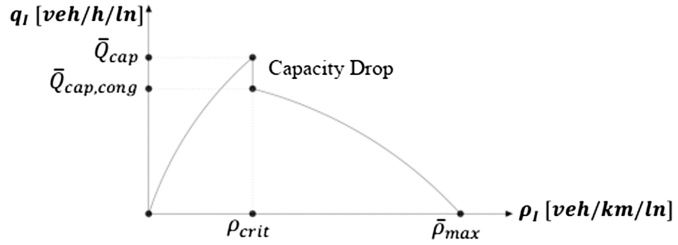


Fig. 3. Sketch of the flow-density fundamental diagram (FD) of the freeway lane drop segment.

built-in functions via VISSIM-COM. In METANET, however, the average number of stops per vehicle cannot be directly computed since the number of stops is a microscopic characteristic and METANET is a macroscopic model. But since the average number of stops is directly related to the extent or the length of the queue that forms upstream of the lane drop, as in the current problem, we change the objective when using METANET to minimize the RMSE between the values of the density of Segment I (which is S10 in our case study, or the last segment upstream the lane drop) *and* the critical density downstream the lane drop [Eq. (10)]. By doing this, the algorithm attempts to prevent traffic flow from transitioning into the congested regime (i.e., try to maintain the traffic state within the free-flow regime for as long as possible).

In that regard, it should be noted that one need not be concerned about the case when the traffic density is equal to zero (in that case, the RMSE value would be high, although the number of stops would be equal to zero with no traffic present at a value of density equal to 0). The reason this case is not a concern is that control is only exerted or activated during the controlled window of high demand (and not under light traffic conditions). In that case, when the demand exceeds the capacity of the lane drop, a traffic density of zero will not occur.

Fig. 3 displays a sketch of the flow-density fundamental diagram (FD) of the freeway lane drop segment. Further details and analysis regarding the FD and the capacity drop phenomenon can be found in Dervisoglu et al. (2009) and Turner (2011). Regarding the other two constrained objectives (i.e., the travel time and vehicle throughput), they can be derived from METANET output as shown in Eq. (10)

$$\min \text{RMSE} = \sqrt{\frac{\sum_{k=Cs}^{Ce} (\rho_I(k) - \rho_{crit} \cdot \lambda_{I+1}/\lambda_I)^2}{K \cdot T}} \tag{10}$$

$$subject\ to \sum_{k=Cs}^{Ce} \sum_{i=1}^{I} \rho_i(k) \cdot L_i \cdot \lambda_i \cdot T \le \mathbb{Z}_2 \cdot f_2^{base,METANET}$$

$$\sum_{k=Cs}^{Ce} q_I(k) \cdot T \ge \mathbb{Z}_3 \cdot f_3^{base,METANET}$$

where λ_I and λ_{I+1} = number of lanes of the last segment (S10) upstream the lane drop, and the segment downstream the lane drop, respectively; ρ_I and q_I = density and exiting flow of the last Segment I (S10), respectively; $\mathbb{Z}_2$ and $\mathbb{Z}_3$ = predefined percentage values; and $f_2^{\text{base,METANET}}$ and $f_3^{\text{base,METANET}}$ = corresponding measures obtained from the base case via the METANET model for travel time and the vehicle throughput, respectively.

Case 2: Variable *Link-Based* Speed Limits Control Strategy

The traditional control strategy of controlling speed limits on a link basis was used as a benchmark against which to compare and gauge



Fig. 4. Variable lane group–based speed limits control strategy with no right lane change allowed (Case 3).

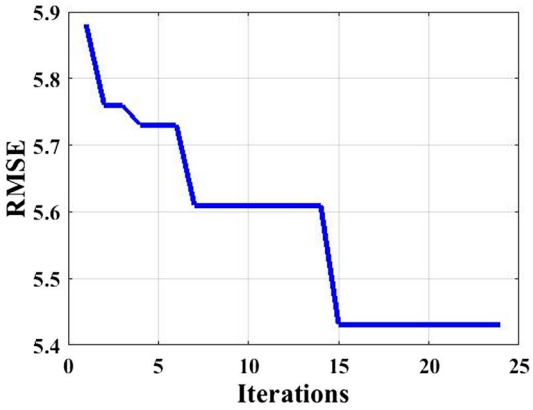


Fig. 5. PSO method convergence during the calibration process of METANET.

the proposed *lane group–based* control strategy. The same steps of the control strategy in Case 1 were followed herein, except that at the same position, the value of speed limits for all lanes was set to be the same (i.e., we do not distinguish between Groups 1 and 2 as previously discussed).

Case 3: Variable *Lane Group–Based* Speed Limits Control Strategy with No Right Lane Change Allowed

The purpose of this case is to show the impact of combining variable lane group–based speed limits with a form of lane control that involves prohibiting right lane changes (i.e., vehicles are only allowed to move into the lane to their left). Fig. 4 depicts the directions of the allowable lane changes (represented by arrows) in the current control strategy. Although this case may not be easily implementable, given that vehicles today are still human-driven, it is easily implementable within the context of self-driving and connected vehicles expected in the not too distant future (Harding et al. 2014; Bagloee et al. 2016; Litman 2018).

Results

This section will first present the results of the METANET model calibration and validation process. This will be followed by a presentation of the findings from the evaluation of the proposed control strategies previously described.

METANET Model Calibration and Validation Outcomes

The PSO calibration of the METANET model yielded the following optimal values for the METANET model parameters a, τ, θ, ψ, ϕ, $\bar{Q}_{cap}$, and $\bar{\rho}_{max}$: 2.13, 0.01 h, 99.69 km^2/h, 13.58 veh/km/ln, 2.00, 2476.25 veh/h/ln, and 131.24 veh/km/ln, respectively. Fig. 5 shows the change in the value of the objective function value (RMSE) with the PSO iterations, and it can be seen that the objective function converged to a value of 5.4304. Fig. 6 compares the density and speeds obtained from METANET to those of the

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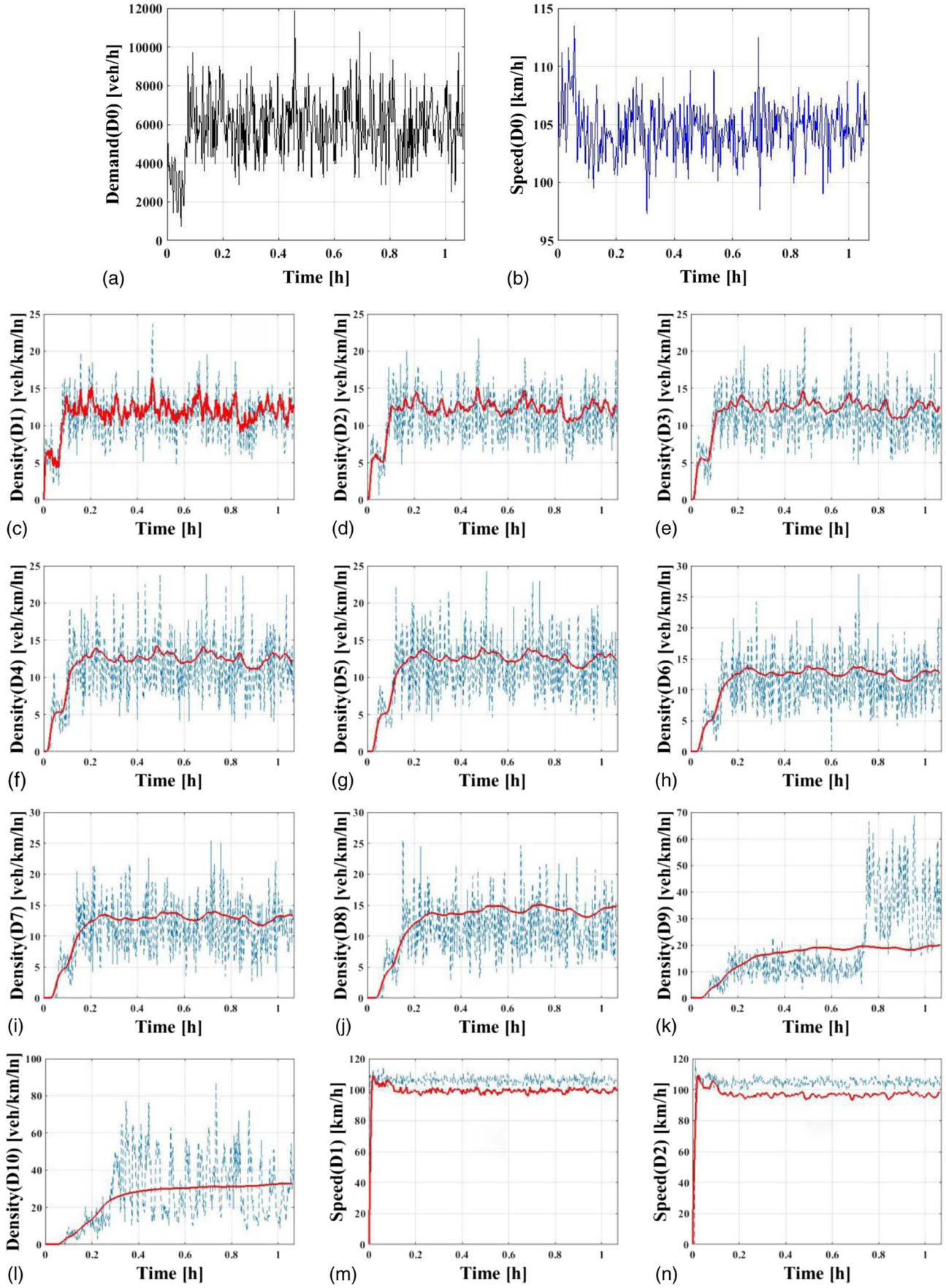


Fig. 6. (a) The input variable of entering demand at D0 for METANET calibration; (b) the input variable of mean speed at D0 for METANET calibration; (c–l) the calibrated density of METANET in solid line versus VISSIM data in dashed line at D1–D10, respectively; and (m–v) the calibrated mean speed of METANET in solid line versus VISSIM data in dashed line at D1–D10, respectively.

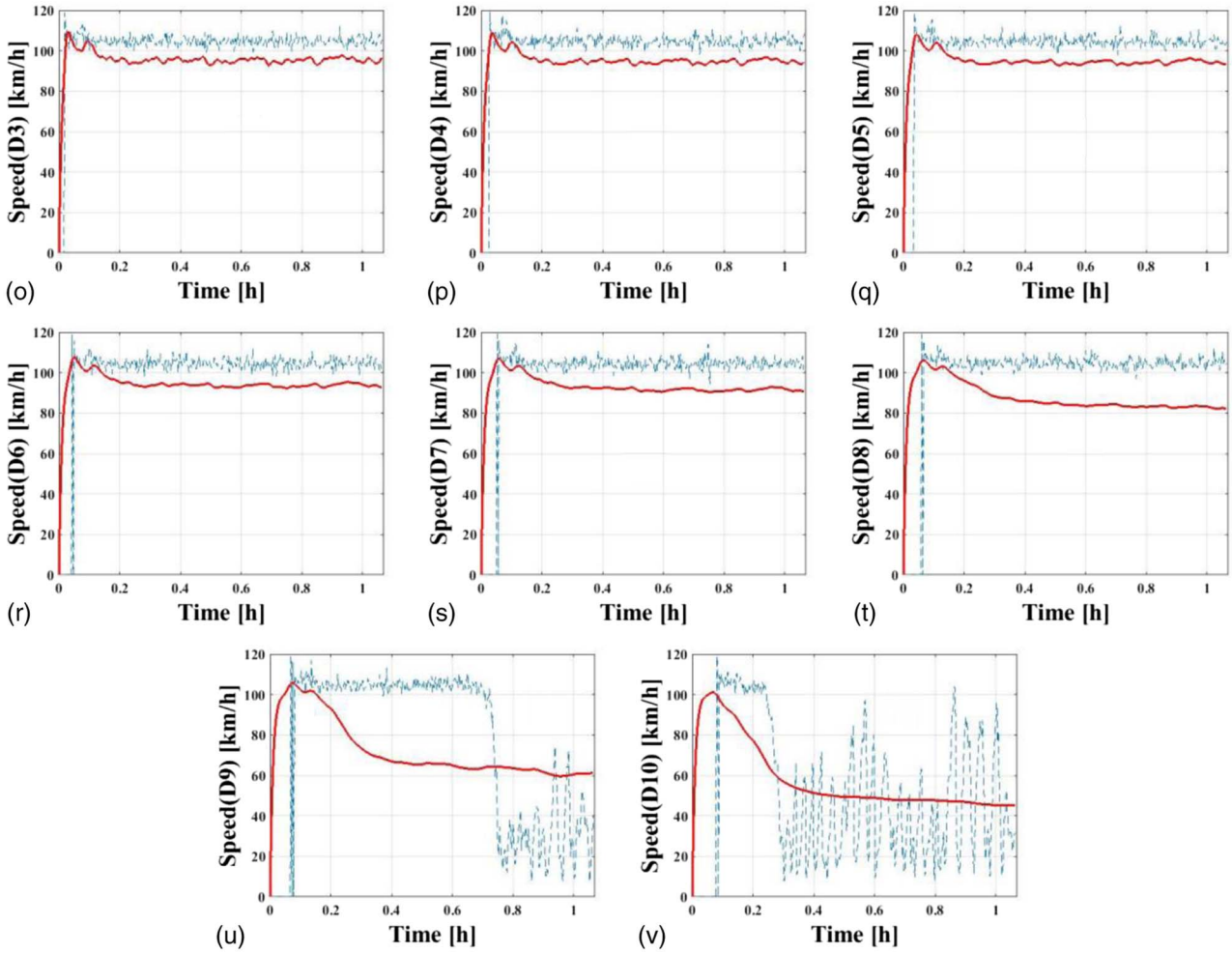


Fig. 6. (*Continued.*)

VISSIM model for the 10 segments [D1–D10], and it can be seen that there is a reasonable match between METANET and VISSIM results, keeping in mind the inherent differences between the two models (note that Figs. 6(a and b) show the input demand and the input speed).

Following the model calibration, the calibrated METANET model was validated by checking the ability of the model to reproduce the traffic dynamics of another scenario different from that which was used in the prior calibration process. For the validation scenario, the vehicle demand in VISSIM was assumed to be equal to 2,700, 5,600, 6,200, 5,200, and 1,600 veh/h, corresponding to the simulation time intervals 0:240, 240:1140, 1140:2040, 2040:2940, and 2940:3840 s, respectively. For that validation scenario, the RMSE was found to be equal to 4.8638. Fig. 7 shows the findings of the METANET model validation by plotting the density and the mean speed obtained from METANET against those of VISSIM for the 10 segments [Figs. 7(a and b) show the input demand and mean speed for the validation scenario]. As can be seen, the METANET model results appear to match the VISSIM output reasonably well.

Traffic Control Strategies Findings

In the present study, the controlled window, extending for 1 h (from the beginning of high demand entrance at time step 240 s to the point in which demand returns back to the low level at time step 3840 s), was discretized into four control intervals, each equal to 15 min. For the base case (i.e., no control), the values of the average number of stops, the average travel time, and the vehicle throughput during the controlled window, were found to be equal to 9.43 stops/veh, 570.92 s, and 4585 veh respectively.

As previously discussed, our multiobjective optimization problem considers three functions $f_1(\boldsymbol{v})$, $f_2(\boldsymbol{v})$, and $f_3(\boldsymbol{v})$, where $f_1(\boldsymbol{v})$ is the main objective function capturing the average number of stops per vehicle, and $f_2(\boldsymbol{v})$ and $f_3(\boldsymbol{v})$ are the two other constrained objectives corresponding to the average travel time and the vehicle throughput, respectively. In general, minimizing the average number of stops per vehicle by controlling the speed limits at freeway lane drops could potentially lead to an increase in the average travel time and a reduction in vehicle throughput. To avoid significant deterioration in the secondary objectives (i.e., travel time and throughput), using the ε–constraint method, $f_2(\boldsymbol{v})$ and $f_3(\boldsymbol{v})$ were constrained to a reduction not to exceed 10% of their values for the base case. In other words, they are constrained as $f_2(\boldsymbol{v}) \leq 1.10 f_2^{\text{base}}$ and $f_3(\boldsymbol{v}) \geq 0.90 f_3^{\text{base}}$.

As previously described, our optimization routine proceeds by first using only the macroscopic traffic model METANET (see the left part of the flowchart in Fig. 1). In that case, the RMSE between the values of the traffic density of Segment I, on the one hand, and the critical density downstream the lane, on the other, is used as the primary objective function. It is worth clarifying that in order to reduce the dimensionality of the optimization problem, the search for the optimal variable speed limit proceeded in increments of 10 km/h, within the upper and lower bounds for the speed limits

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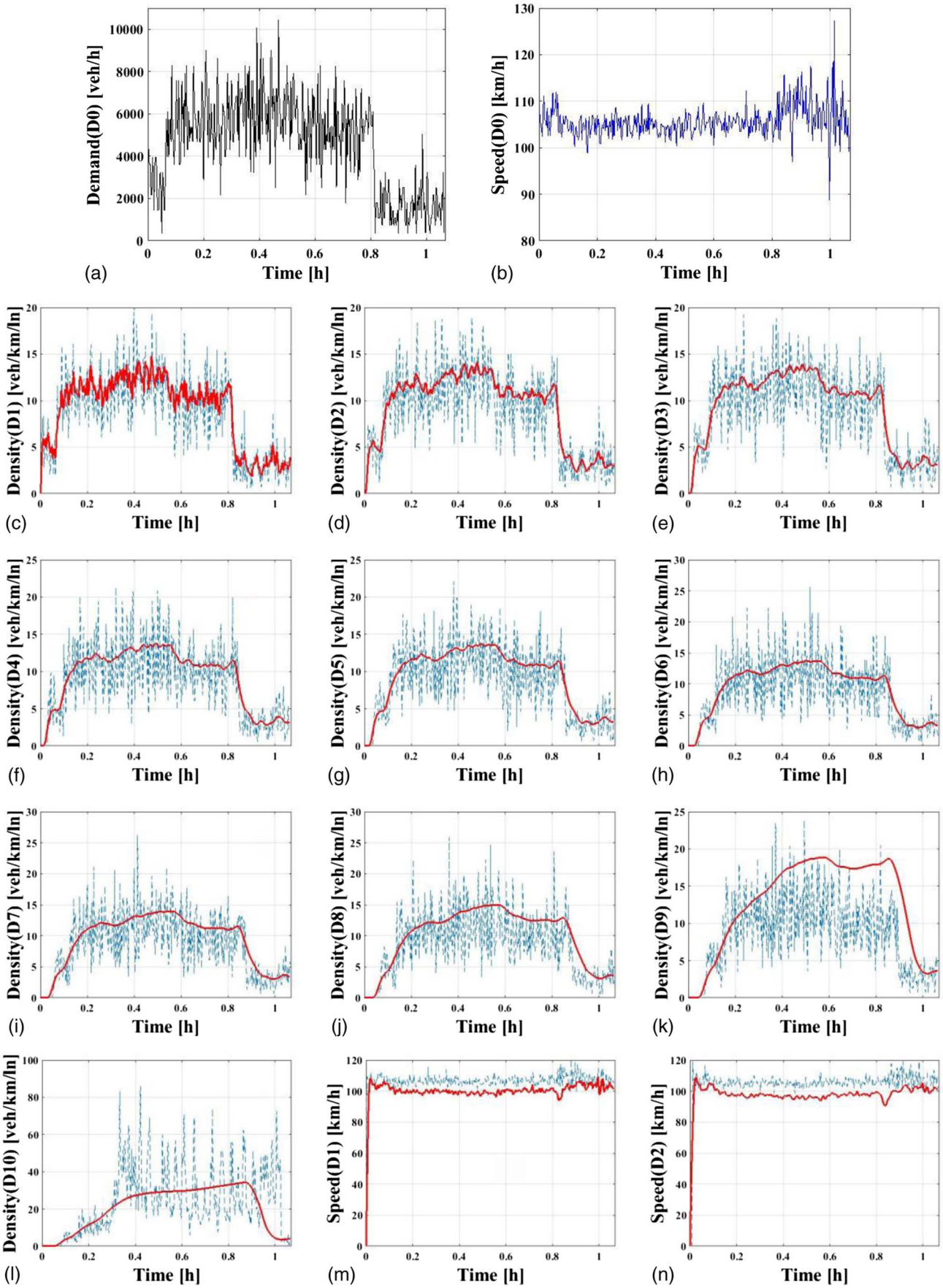


Fig. 7. (a) The input variable of entering demand at D0 for METANET validation; (b) the input variable of mean speed at D0 for METANET validation; (c–l) the validated density of METANET in solid line versus VISSIM data in dashed line at D1–D10, respectively; and (m–v) the validated mean speed of METANET in solid line versus VISSIM data in dashed line at D1–D10, respectively.

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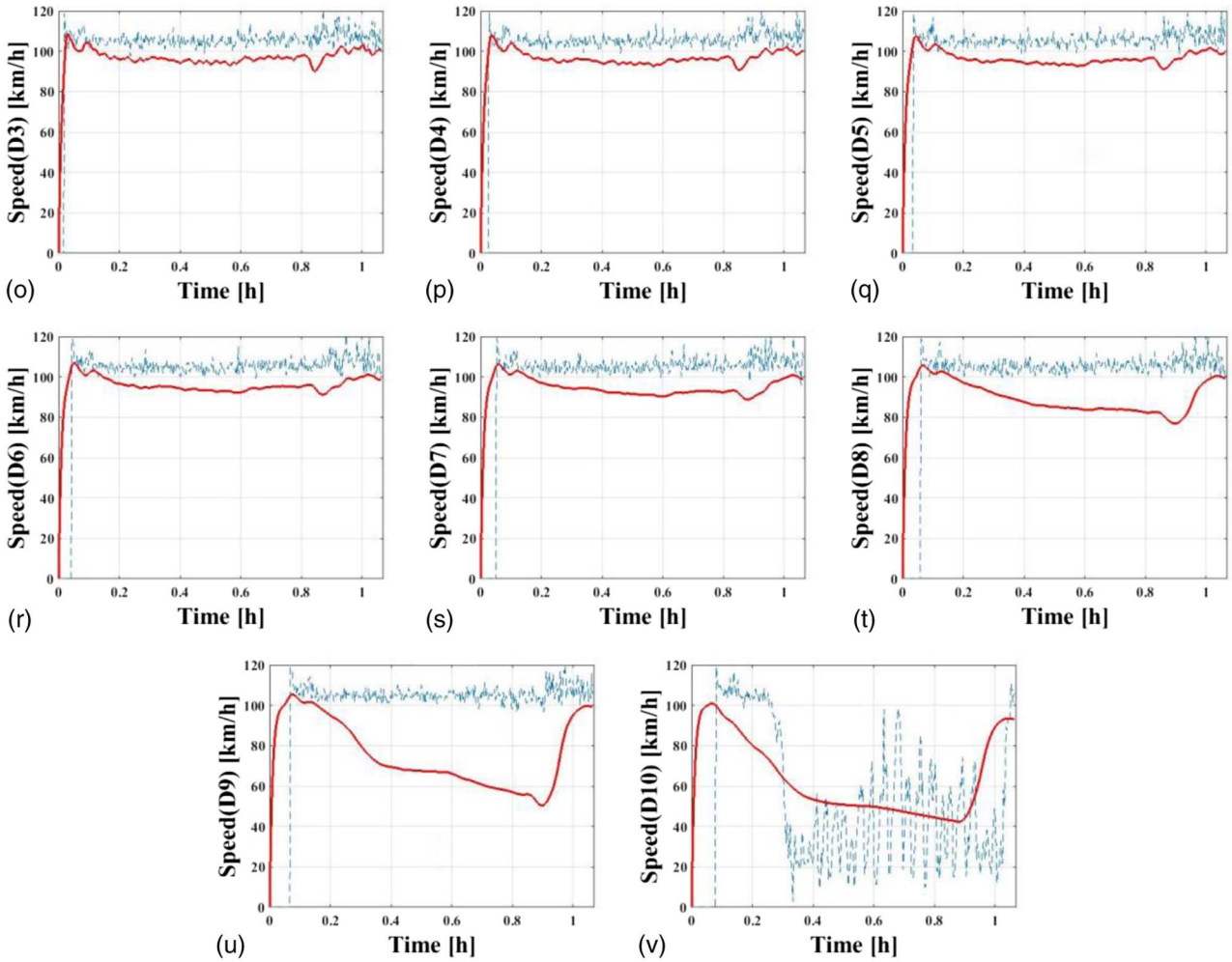


Fig. 7. (*Continued.*)

of V_{free} and 20 km/h, respectively. With respect to the PSO algorithm parameters, the swarm population size was set as 50 agents and the maximum number of iterations was 100. The PSO method ran until the stopping criterion was met; this occurred either when the maximum number of iterations was reached or when 10 consecutive iterations had the same value of the main objective function. The assumption of 10 consecutive iterations was selected because it was found from experiments that it results in good outcomes in terms of engineering applications with keeping a good balance with the consideration of the execution time of optimization. Although it seems strict, it is better, in the current problem, than assuming a minimum error between only two consecutive iterations.

The optimal VSLs determined from the use of METANET for evaluation, in conjunction with PSO, are provided in Table 1 (the METANET case). These values were then used to seed the initial population for the PSO, when used in conjunction with VISSIM, so that the algorithm would not be starting at a random point. Doing this is intended to reduce the required runtime of the algorithm, especially given that running VISSIM is quite demanding in terms of execution time. The METANET case required an execution time of about 5 min, but each of the control Cases 1, 2, and 3 required about three days on the first author's laptop, which has modest specifications [namely an Intel(R) Core(TM) i3-3217U CPU @ 1.80 GHz, and a memory (RAM) of 4.00 GB]. One natural way to speed up the processing is to run the evaluation simultaneously on several workstations in a cluster (in this study, we ran the model on only one workstation, hence, the rather long execution time). The optimal VSLs for control Cases 1, 2, and 3, are also provided in Table 1.

Table 2 provides the corresponding values of the traffic flow performance criteria obtained when optimizing using METANET alone (the METANET case), as well as after using VISSIM as the evaluation function in conjunction with PSO, and after seeding the initial population with the METANET agent (Cases 1, 2, and 3). As can be seen, using METANET optimal speed limits helped in leading to a reduced average number of stops of 8.46 stops/veh, which is about a 10.29% reduction with respect to the base case; this shows that the use of METANET does indeed lead to discovering good start points for the more detailed optimization using VISSIM. The convergence process of the PSO algorithm in that case is shown in Fig. 8(a).

After running PSO again, in conjunction with VISSIM (see the right half of the flowchart in Fig. 1), further improvements in the objective functions were achieved. Specifically, as can be seen from Table 2, the proposed control strategy (Case 1) resulted in a significantly reduced average number of stops per vehicle compared to that found in the base case. The average number of stops value was equal to 4.81 stops/veh, which is around a 48.99% improvement with respect to the base case. The PSO convergence for that case is shown in Fig. 8(b).

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Table 1. Values of optimal speed limits for each case

Case	Control interval	Optimal speed limits values (km/h)									
METANET		U_1		U_2		U_3		U_4		U_5	
	1st	100		100		100		100		100	
	2nd	40		80		60		90		100	
	3rd	90		100		100		100		100	
	4th	100		70		50		100		100	
1		$U_{1\text{-}1}$	$U_{1\text{-}2}$	$U_{2\text{-}1}$	$U_{2\text{-}2}$	$U_{3\text{-}1}$	$U_{3\text{-}2}$	$U_{4\text{-}1}$	$U_{4\text{-}2}$	$U_{5\text{-}1}$	$U_{5\text{-}2}$
	1st	100	100	100	100	100	100	100	100	100	100
	2nd	50	50	70	70	100	70	100	100	100	100
	3rd	30	30	30	30	100	100	100	100	100	100
	4th	60	60	90	80	80	80	80	70	100	100
2		U_1		U_2		U_3		U_4		U_5	
	1st	100		100		100		100		100	
	2nd	50		70		80		100		100	
	3rd	30		30		100		100		100	
	4th	60		90		80		70		100	
3		$U_{1\text{-}1}$	$U_{1\text{-}2}$	$U_{2\text{-}1}$	$U_{2\text{-}2}$	$U_{3\text{-}1}$	$U_{3\text{-}2}$	$U_{4\text{-}1}$	$U_{4\text{-}2}$	$U_{5\text{-}1}$	$U_{5\text{-}2}$
	1st	100	100	100	100	100	100	100	100	100	100
	2nd	50	50	70	70	70	70	100	100	100	100
	3rd	20	30	50	50	100	100	100	100	100	100
	4th	60	60	70	80	80	80	100	100	100	90

The traditional control strategy of variable link-based speed limits (Case 2) resulted in a value equal to 5.00 stops/veh, which is a little greater than the value obtained by Case 1 (Table 2). Nevertheless, this value represents a 46.98% reduction with respect to the base case. The PSO convergence for Case 2 is shown in Fig. 8(c).

Finally, the impact of the combination of variable lane group–based speed limits and no right lane change on the average number of stops (Case 3) was found to be quite favorable. Specifically, this strategy reduced the average number of stops to 4.59 stops/veh, which is around a 51.33% improvement with respect to the base case (the greatest reduction out of the three control strategies tested). The PSO convergence for Case 3 is shown in Fig. 8(d).

It is also to be noted that, in all cases, the corresponding increase in the average travel time was less than the 10% bound imposed on the problem. Moreover, the vehicle throughout does not appear to be negatively impacted to any significant level. In fact, for Case 3, there was even an increase of about 1.42% in throughput with respect to the base case.

It must be mentioned that our goal in conducting these experiments and evaluation on the hypothetical problem was to serve as a proof-of-concept for the ability of the proposed control strategy to improve traffic flow through its ability to reduce the average number of stops. The actual magnitude of the reduction (which is around 50% in our case) is likely going to vary from one scenario to the next, according to several factors. Those factors include such things as the site characteristics, geometry, traffic flow situation, input demand, surrounding environment, and prespecified traffic control restrictions.

Table 2. Values of traffic flow performance criteria for each case via VISSIM

Case	Average number of stops (stops/veh)	Average travel time (s)	Vehicle throughput (veh)
Base	9.43	570.92	4,585
METANET	8.46 (↓10.29%)	608.29 (↑6.55%)	4,573 (↓0.26%)
1	4.81 (↓48.99%)	611.09 (↑7.04%)	4,581 (↓0.09%)
2	5.00 (↓46.98%)	617.55 (↑8.17%)	4,578 (↓0.15%)
3	4.59 (↓51.33%)	625.69 (↑9.59%)	4,650 (↑1.42%)

To get a feel for how the magnitude of the reduction in the number of stops would vary with some of the aforementioned factors, additional experiments with different values of the input demand were performed. Table 3 provides the new assumed values of the input demand and summarizes the corresponding results of the traffic performance criteria. Over the simulation time horizon of 0:240–240:3840 s, the new assumed input demands were set as 3,000–5,000, 3,000–7,000, and 3,000–8,000 veh/h within three different additional experiments (instead of 3,000–6,000 veh/h that was previously studied). It is to be noted that the value of 3,000 veh/h is considered as a warm-up initialization and that is why it was not changed. The experiments were performed using the variable *lane group–based* speed limits control strategy only because it is the one of interest, the high costly execution time of the optimization process required, and for the sake of brevity. As presented in Table 3, the results indicate that the proposed approach managed to minimize the average number of stops by up to around 21% when the input demand was 5,000 veh/h, and up to around 55% and 47% for the cases of 7,000 and 8,000 veh/h, respectively, when compared to the average number of stops for the base case. This confirms that the proposed approach is likely to yield more significant benefits in heavy congested traffic flow situations at freeway lane drop areas, compared to cases of light traffic.

Conclusions

The current research presented a multiobjective, optimal, lane group–based, VSL traffic control strategy via a joint modeling approach. The joint approach consisted of running a calibrated and validated macroscopic model, METANET, first to derive initial solutions fast, followed by running the microscopic traffic simulation model VISSIM. The PSO technique and the ε-constraint method were combined in the optimization process. The impacts on both traffic safety and efficiency of the proposed control strategy at freeway lane drop areas were investigated on a hypothetical freeway lane drop area under a life-like congested traffic scenario.

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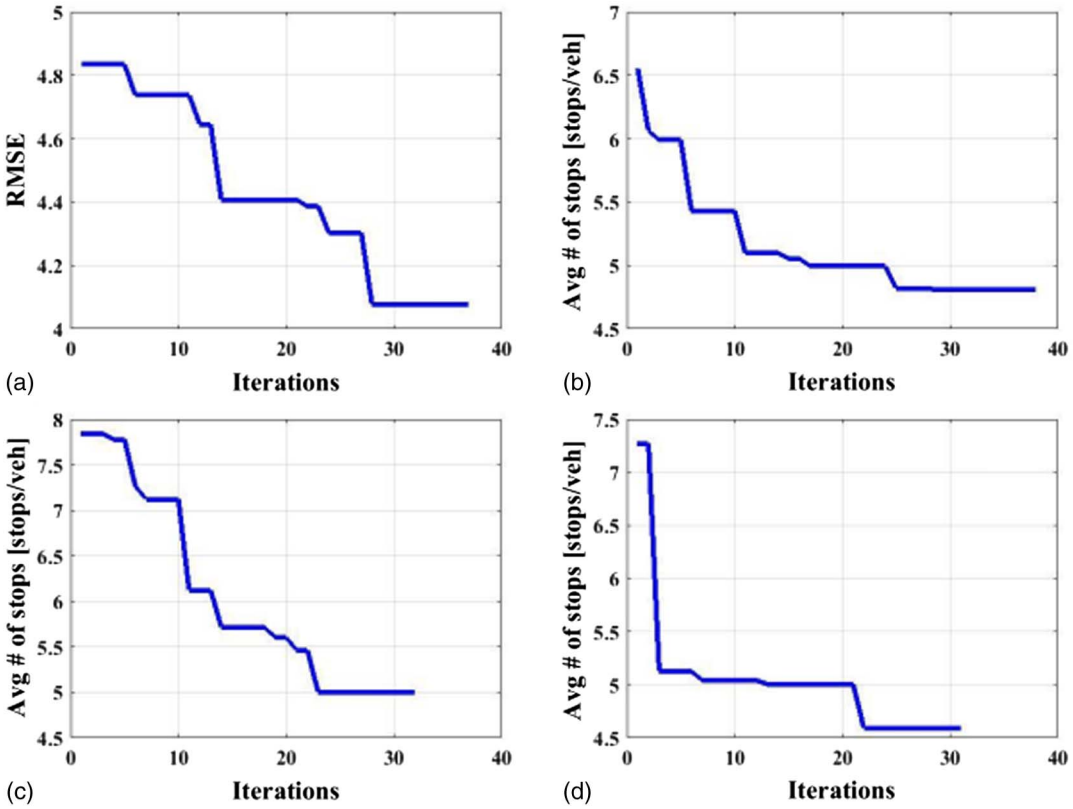


Fig. 8. The PSO method convergence during the estimation process of (a) METANET agent; (b) Case 1 agent; (c) Case 2 agent; and (d) Case 3 agent.

Table 3. Values of traffic flow performance criteria corresponding to additional experiments with different input demands using variable lane group–based speed limits control strategy via VISSIM

Demand (veh/h)	Simulation time (s)	Case	Average number of stops (stops/veh)	Average travel time (s)	Vehicle throughput (veh)
3,000–5,000	0:240-240:3840	Base	3.99	443.87	4,447
		Control	3.15 (↓21.05%)	481.30 (↑8.43%)	4,190 (↓5.78%)
3,000–7,000	0:240-240:3840	Base	16.51	707.71	4,639
		Control	7.46 (↓54.82%)	772.85 (↑9.20%)	4,657 (↑0.39%)
3,000–8,000	0:240-240:3840	Base	21.46	809.28	4,610
		Control	11.37 (↓47.02%)	881.89 (↑8.97%)	4,351 (↓5.62%)

The research findings show that the proposed traffic control strategy is capable of leading to a significant reduction in the average number of stops per vehicle. The improvement reached up to nearly 50% with respect to the base case during the congested traffic situations. This was achievable while avoiding significant deteriorations in the values of other traffic performance criteria such as average travel time and vehicle throughput (the reductions were constrained in our study to at most 10% difference with respect to the base case). The comparison between the proposed traffic control strategy and the traditional variable link-based speed limits showed that the proposed VSLs on a *lane-group basis* lead to a further reduction in the average number of stops. The impact of the combination of variable lane group–based speed limits and no right lane change allowed resulted in the greatest reduction in the average number of stops per vehicle value.

Given the rather long run time required for VISSIM evaluations, the proposed control strategy is more suited for off-line traffic control management for recurrent congested traffic situations (e.g., daily peak hours). In order to implement it in online applications, a cluster of workstations is required to run the model in parallel in order to reduce the run time of the computationally demanding VISSIM-based optimization process. Another alternative is to use the proposed framework in a quasi-real time fashion by developing a library of cases off-line. Finally, it is to be noted that since the proposed traffic control strategy is based on a lane-group basis, it can be extended for future control strategies that will include self-driving or connected vehicles, in which the location of the optimal VSLs can be replaced by the positions of the vehicles receiving the optimal speed advisories.

Data Availability Statement

Some or all data, models, or code generated or used during the study are available from the corresponding author by request.

- Traffic input data
- VISSIM simulation model

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